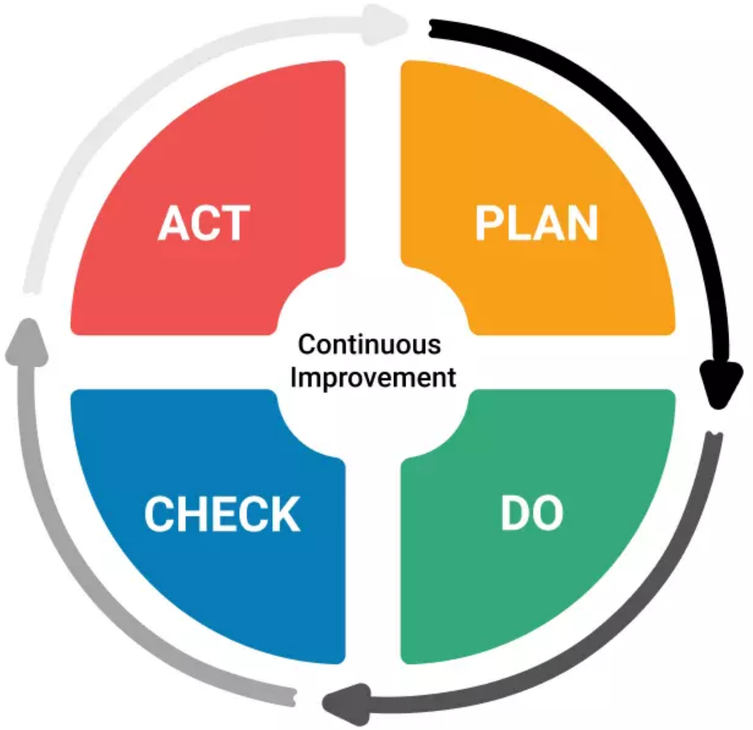
Learning to Learn

Part 2: Group Learning

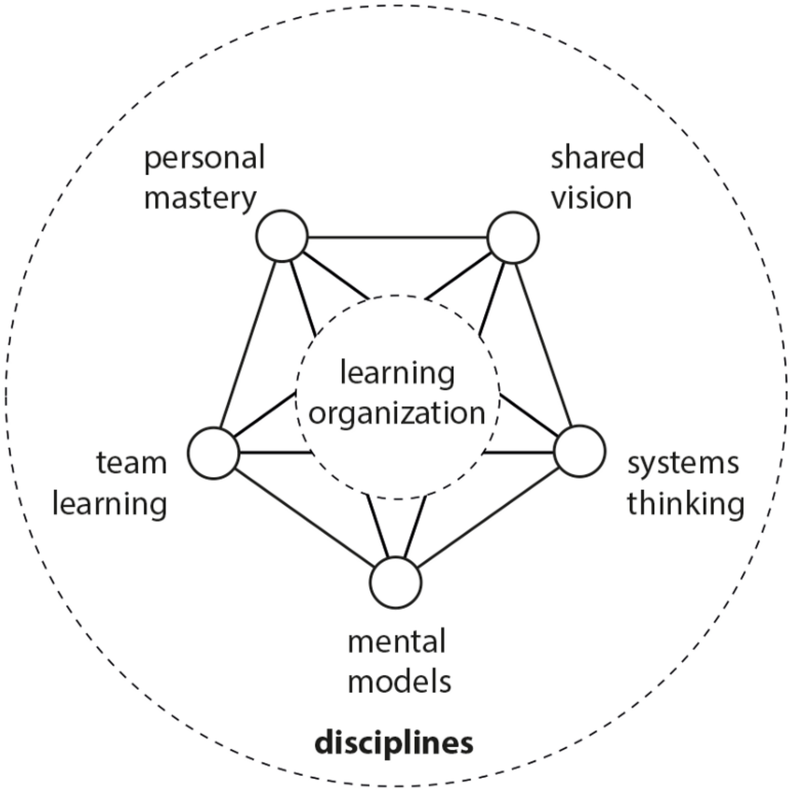
History of organizational learning

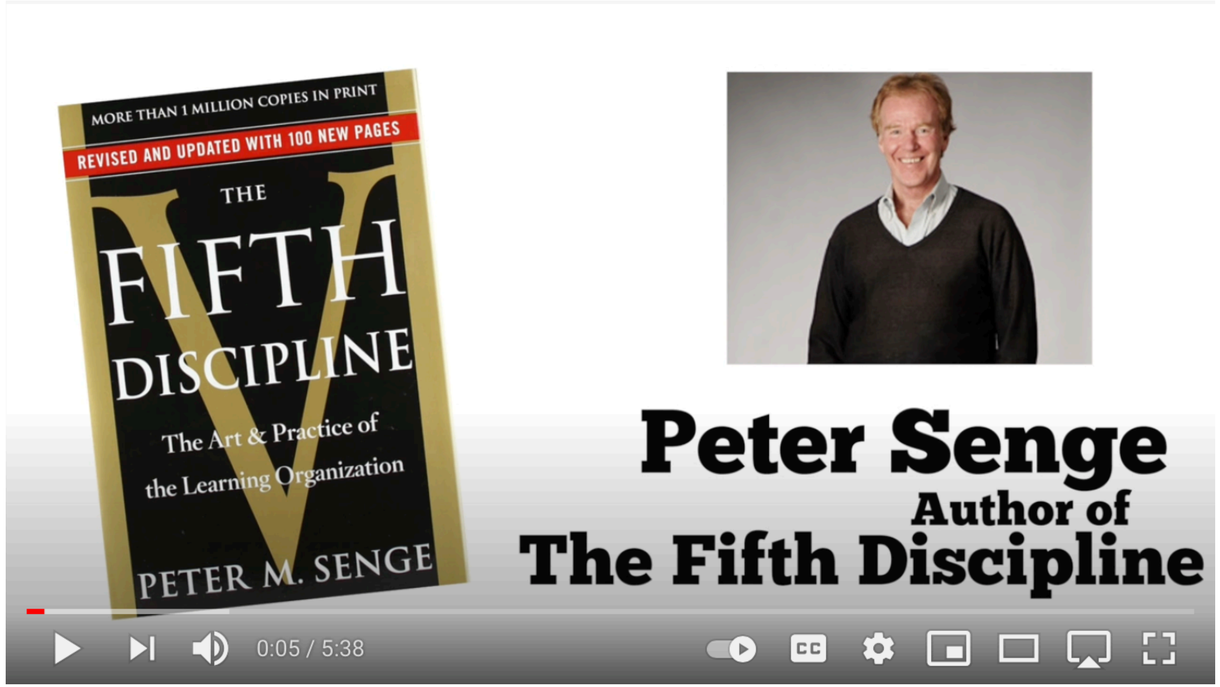
- W. Edwards Deming
  - Continuous improvement (kaizen)
  - Total quality management



Sources: Image from "What is Plan-Do-Check-Act Cycle?", Kanbanize

Organizational Learning - 5 Disciplines

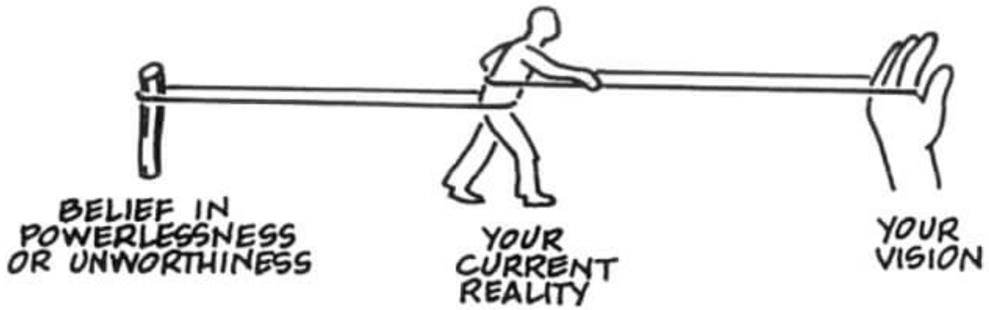




Sources: "The Fifth Discipline", Peter Senge; Image from "The Age of the Learning Organization Is About to Begin", Denniis Hambeukers

Organizational Learning - Aspirations

- Personal Mastery
  - Clarifying our personal vision
  - Striving to see reality clearly
  - Tension between reality and vision
- Shared Vision
  - Finding commonalities of purpose
  - Cannot be solely dictated top-down

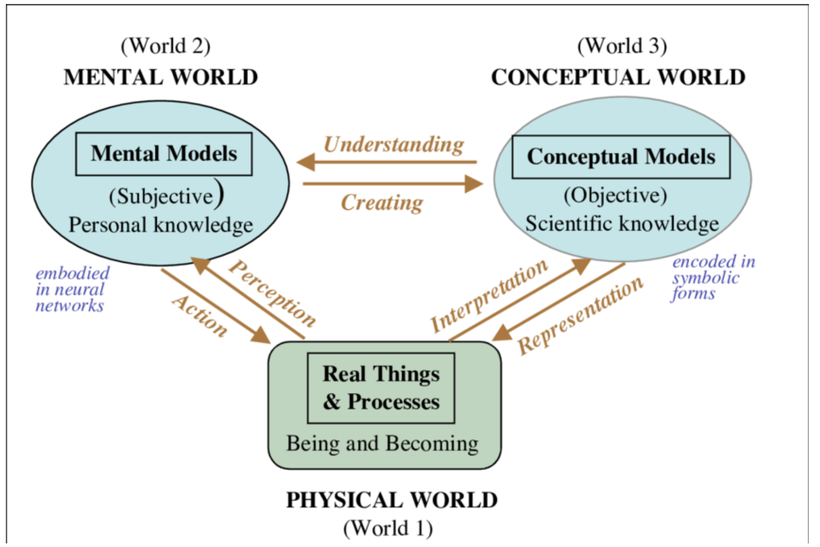


Credit: Peter M Senge.. The Fifth Discipline pg 157

Source: “The Fifth Discipline”, Peter Senge

Organizational Learning - Reflective Conversation

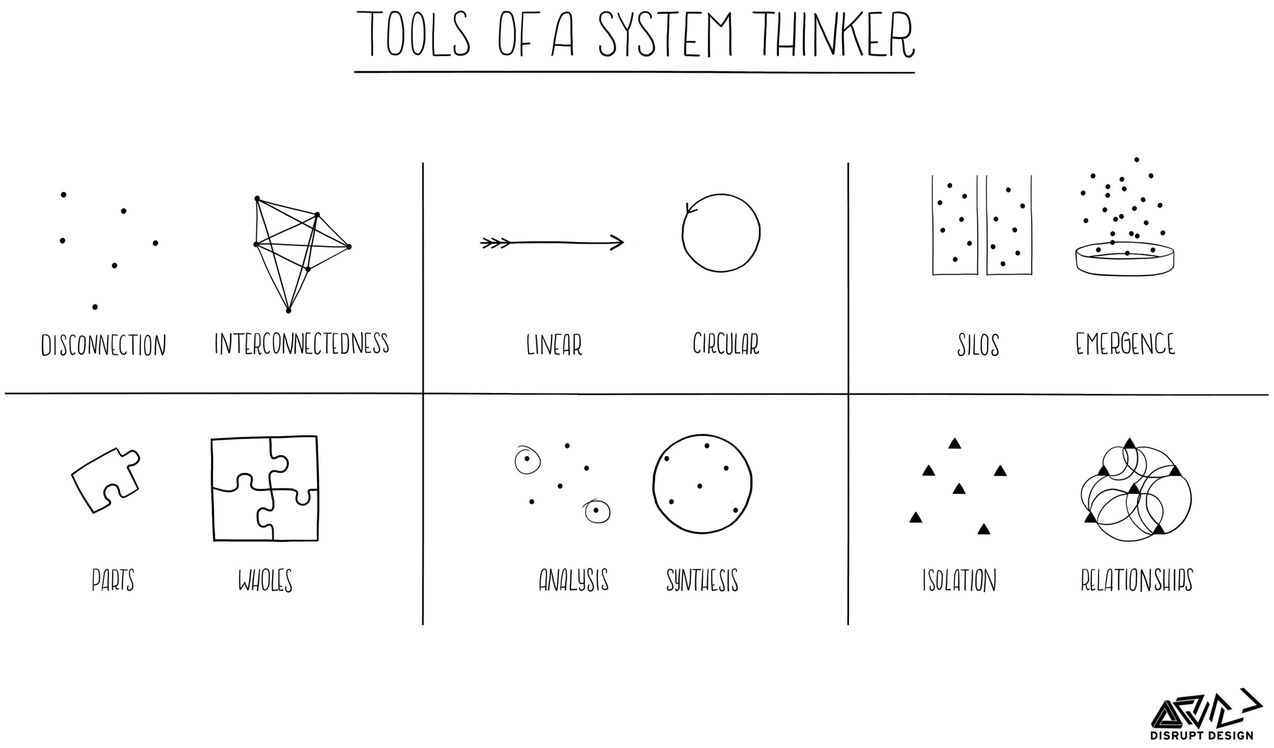
- Mental Models
  - Understanding our assumptions
  - Not always well understood
  - Bringing together different perspectives
  - Tool: Left-hand column

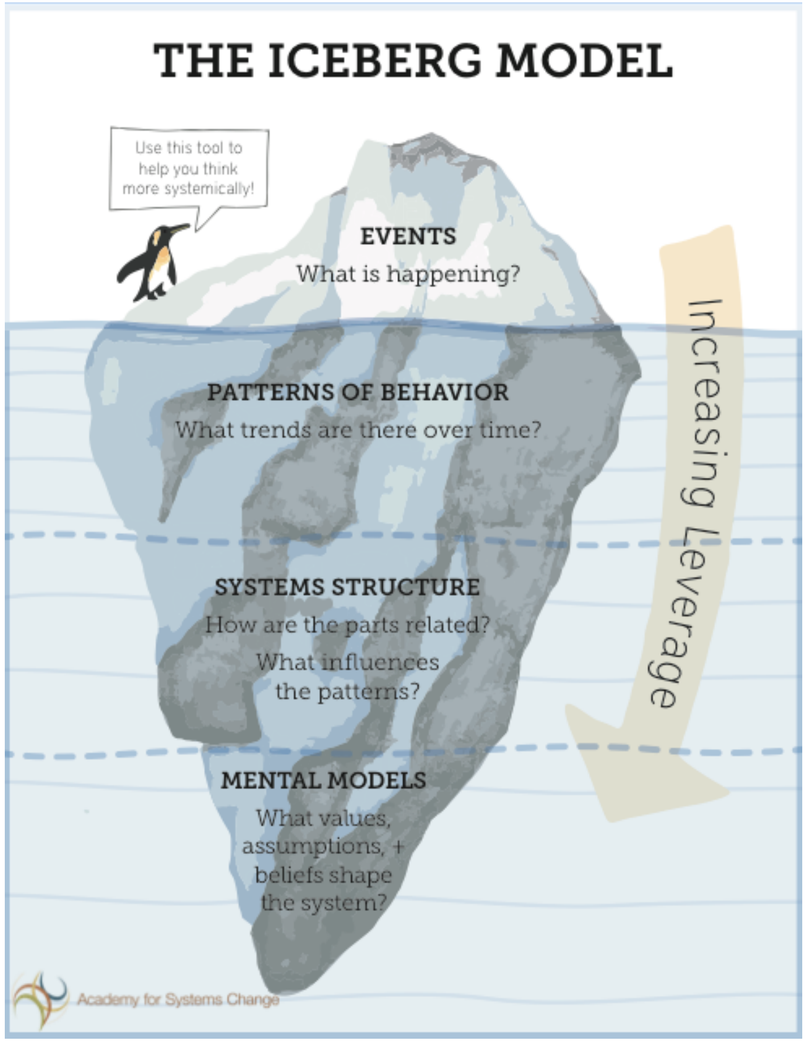


- Team Learning
  - Dialogue - Open-ended conversation to understand perspectives (Diverging)
  - Discussion - Making decisions (Converging)
  - Positive conflict

Source: "The Fifth Discipline", Peter Senge; image from "Notes for a Modeling Theory of Science, Cognition and Instruction", David Hestenes

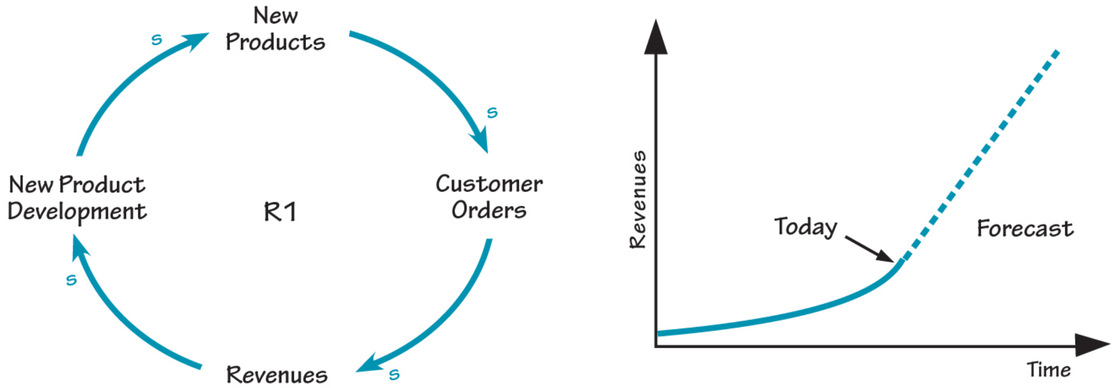
Systems Thinking



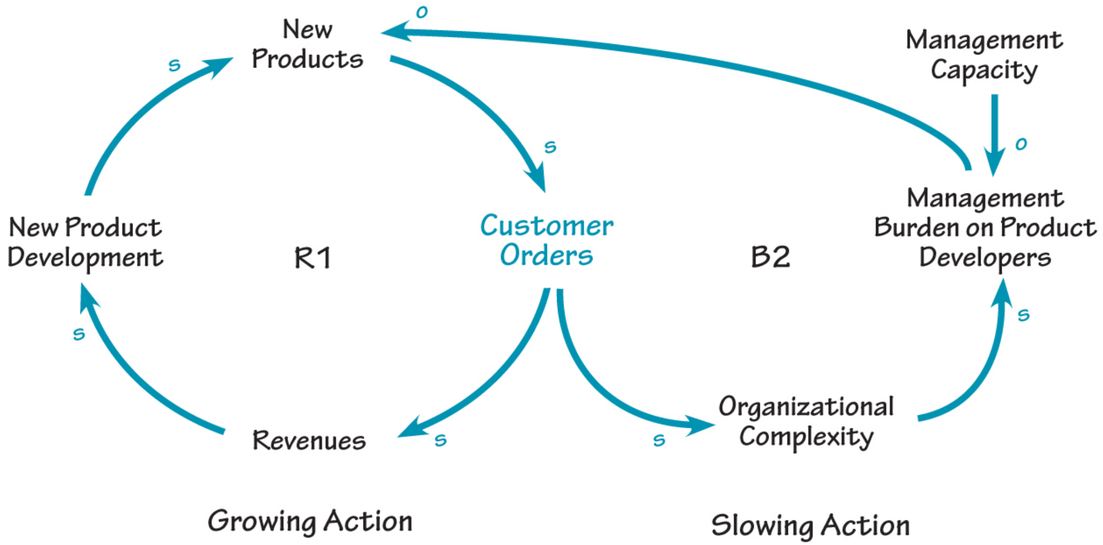


Source: Disrupt Design; [The Donella Meadows Project]

Systems Thinking: Example



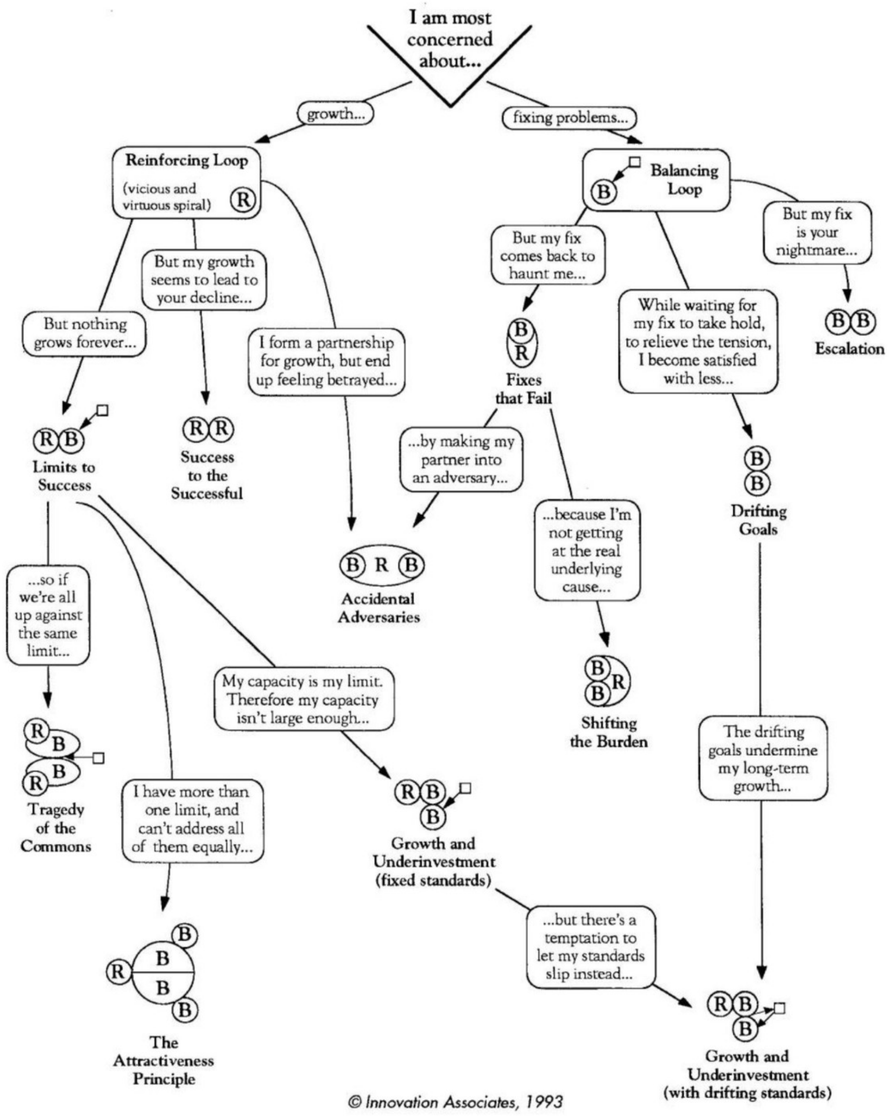
Planned

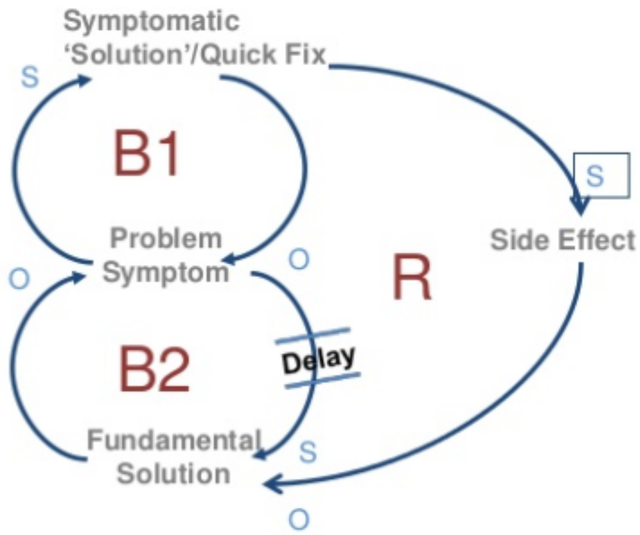


Reality

Source: "The Systems Thinker", Daniel Kim

Systems Thinking: Archetypes





"Shifting the Burden"

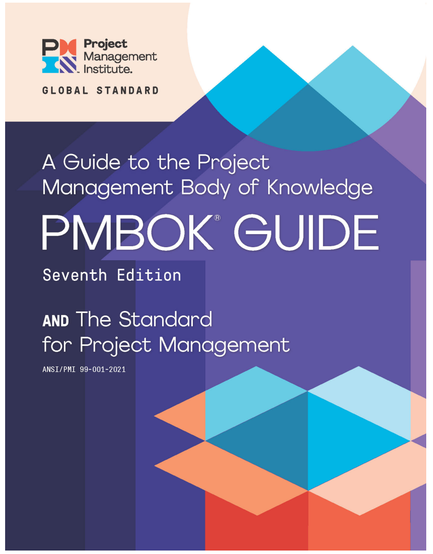
Source: "Using System Archetypes to Shortcut Problem Solving", Tom Connor

Systems Thinking and Group Learning: Further Resources

- Donella (Dana) Meadows, *Thinking in Systems*
- The Donella Meadows Project
- The Systems Thinker
- Introduction to Systems Archetypes
- Peter Senge, *The Fifth Discipline*
- *The Fifth Discipline Fieldbook*
- J. Alex Sherrer, “A Project Manager’s Guide to Systems Thinking” (Part 1) (Part 2)
- Dennis Hambeukers, “The Age of the Learning Organization is About to Begin”
- Reiner Albert Schneider, “Managing projects in complex business environments”
- Tyler La, “Why designers should find the balance between systems thinking and design thinking”
- PMI & Queensland University of Technology, “Training manual for managing complexities and ambiguities in projects”

PMBOK Guide - 6th vs 7th Edition

- 6th Edition
  - Lessons learned register
  - Resource Management
  - Emphasis on processes, tools and techniques
- 7th Edition
  - Project Work Performance Domain - Learning Throughout the Project
  - Principle - Create a collaborative team environment
  - Method - Meetings for lessons learned and retrospectives



Lean principles

- Respect for people
- Continuous improvement
- Amplify learning
- Empower the team
- Optimize the whole

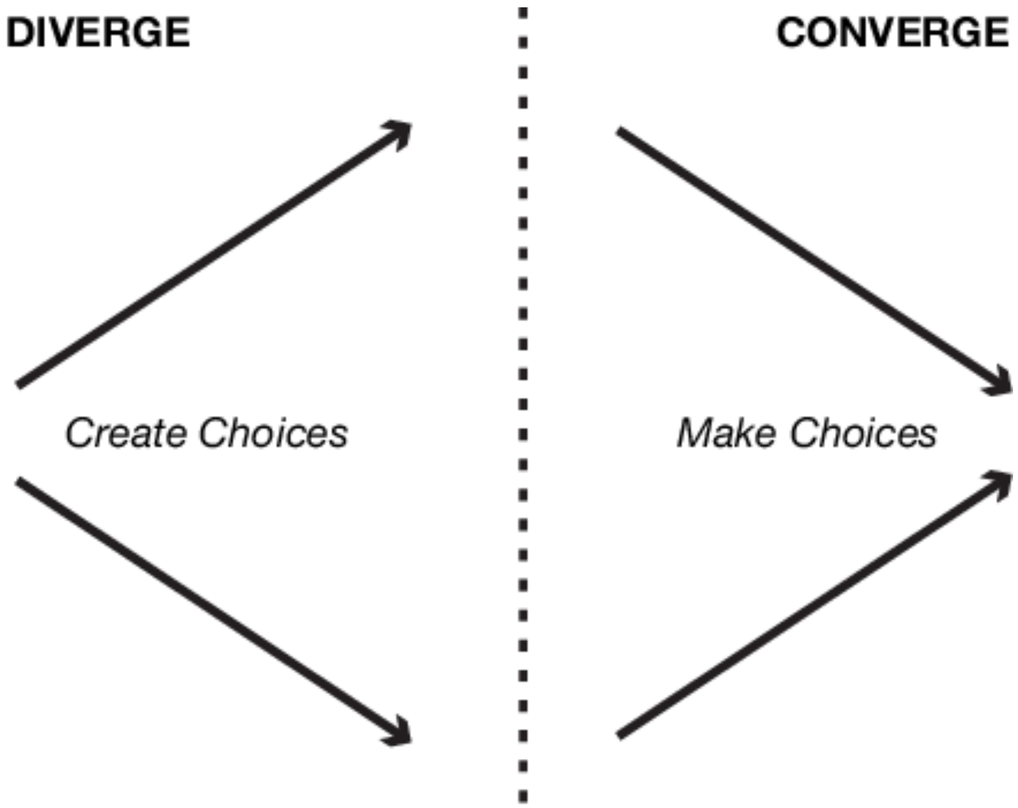
Agile

- Retrospectives
- Cross-functional - frequent business analyst and stakeholder input
- Self-organizing teams

DevOps

- Continuous experimentation and learning
- CAMS - **Culture**, Automation, Measurement, **Sharing**
- Cross-functional - Breaking down barriers and silos
- Problems can be identified quicker

Design Thinking



- Brainstorming
- Interviewing
- Observing
- Prototyping
- Testing

Source: Image from "Understanding the Creative Mechanisms of Design Thinking: An Evolutionary Approach", Katja Thoring

Other methods and artifacts for learning about problems

- Fishbone diagram
- Five why’s
- Root cause analysis
- Questionnaires and surveys
- Flowcharts
- Affinity diagrams

Group Learning - Key Takeaways

- Understand different perspectives and assumptions
- Have honest, exploratory conversations to learn from each other
- Different approaches have tools and techniques to help us do this
- With complex problems, think of the whole and the connections between things (integration) in order to find leverage
- Culture is key